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Quantifying Anomalous Heat Flow under Trade-Off Enhancement Dynamics

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E-mail: skyshn95@unist.ac.kr**Keywords:** Quantum thermodynamics, Quantum correlation, master equation, Quantum information

Abstract

Studies about quantum thermodynamics in the information-theoretic aspects bring out how Information as its quantum nature, consumed as correlation, can change the direction of energy flow from hot to cold, which is called anomalous energy flow(AEF). Here, we study the bounds of AEF in the system-catalyst model by adding the symmetrical constraint for the quantum map, which can reinforce the previous result by adding the dynamical point of view. Our result shows that by constraining the symmetric property of spending information, the maximal bound of AEF increases. We show our result in the coherent catalyst case, which, depending on the Caldeira-Leggett model, can be reproduced using the quantum optics method.

1 Introduction

The quantum correlation overcomes the restriction of the classical limit of heat flow and the opposite direction of heat flow. However, to overcome certain selection rules, adopting the auxiliary system is necessary, which occurs when the system turns out to be open rather than an energy-conserved case.

1.1 Subsection title

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1.1.1 Subsubsection heading

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Figure 1. Text describing the figure and the main conclusions drawn from it. To make your figures accessible to as many readers as possible, try to avoid using colour as the only means of conveying information. For example, in charts and graphs use different line styles and symbols. Further information is available in the online guide: <https://publishingsupport.iopscience.iop.org/publishing-support/authors/authoring-for-journals/writing-journal-article/#figures>

Table 1. Caption text describing the table. Adapt the template table below or replace with a new table. To add more tables, copy and paste the whole `\begin{table}...\end{table}` block.

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2 Results

Acknowledgments

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Funding

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Data availability

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Supplementary data

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References